Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.42)

Metric

$$ds^2 = \frac{a^2}{(t^2-1)F^2}\mathrm{d}t \otimes \mathrm{d}t + \left(\frac{a^2}{x^2F^2+F^2}\right)\mathrm{d}x \otimes \mathrm{d}x + \left(-\frac{a^2t^2-a^2}{F^2}\right)\mathrm{d}y \otimes \mathrm{d}y + \left(\frac{a^2x^2+a^2}{F^2}\right)\mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$(x^\mu) = (t, x, y, z)$

Metric Functions

$F(t, x) = F$

Null Tetrad

Covariant components

$$l = \left(\frac{a^2}{2\sqrt{-t^2+1}F^2}\right)\,dt + \left(\frac{a^2}{2\sqrt{x^2+1}F^2}\right)\,dx$$

$$n = \left(\frac{1}{\sqrt{-t^2+1}}\right)\,dt + \left(-\frac{1}{\sqrt{x^2+1}}\right)\,dx$$

$$m = \left(\frac{i\sqrt{-\frac{1}{2}\,t^2+\frac{1}{2}}a}{F}\right)\,dy + \left(\frac{\sqrt{\frac{1}{2}\,x^2+\frac{1}{2}}a}{F}\right)\,dz$$

$$\overline{m} = \left(-\frac{i\sqrt{-\frac{1}{2}\,t^2+\frac{1}{2}}a}{F}\right)\,dy + \left(\frac{\sqrt{\frac{1}{2}\,x^2+\frac{1}{2}}a}{F}\right)\,dz$$

Contravariant components

$$l = \left(\frac{t^2}{2\sqrt{t+1}\sqrt{t-1}} - \frac{1}{2\sqrt{t+1}\sqrt{t-1}}\right) dt + \left(\frac{1}{2}\sqrt{x^2+1}\right) dx$$

$$n = \left(\frac{t^2F^2}{a^2\sqrt{t+1}\sqrt{t-1}} - \frac{F^2}{a^2\sqrt{t+1}\sqrt{t-1}}\right) dt + \left(-\frac{x^2F^2}{\sqrt{x^2+1}a^2} - \frac{F^2}{\sqrt{x^2+1}a^2}\right) dx$$

$$m = \left(-\frac{i\sqrt{t+1}\sqrt{t-1}F}{\sqrt{2}at^2 - \sqrt{2}a}\right) dy + \left(\frac{\sqrt{x^2+1}F}{\sqrt{2}ax^2 + \sqrt{2}a}\right) dz$$

$$\overline{m} = \left(\frac{i\sqrt{t+1}\sqrt{t-1}F}{\sqrt{2}at^2 - \sqrt{2}a}\right) dy + \left(\frac{\sqrt{x^2+1}F}{\sqrt{2}ax^2 + \sqrt{2}a}\right) dz$$

Directional Derivatives

$$D = \frac{t^2}{2\sqrt{t+1}\sqrt{t-1}} - \frac{1}{2\sqrt{t+1}\sqrt{t-1}}\,\partial_t + \frac{1}{2}\sqrt{x^2+1}\,\partial_x$$

$$\Delta = \frac{t^2F^2}{a^2\sqrt{t+1}\sqrt{t-1}} - \frac{F^2}{a^2\sqrt{t+1}\sqrt{t-1}}\,\partial_t - \frac{x^2F^2}{\sqrt{x^2+1}a^2} - \frac{F^2}{\sqrt{x^2+1}a^2}\,\partial_x$$

$$\delta = -\frac{i\sqrt{t+1}\sqrt{t-1}F}{\sqrt{2}at^2 - \sqrt{2}a}\,\partial_y + \frac{\sqrt{x^2+1}F}{\sqrt{2}ax^2 + \sqrt{2}a}\,\partial_z$$

$$\overline{\delta} = \frac{i\sqrt{t+1}\sqrt{t-1}F}{\sqrt{2}at^2 - \sqrt{2}a}\,\partial_y + \frac{\sqrt{x^2+1}F}{\sqrt{2}ax^2 + \sqrt{2}a}\,\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{t+1}\sqrt{t-1}tx^2}{4\left(t^2x^2+t^2-x^2-1\right)} - \frac{\sqrt{x^2+1}t^2x}{4\left(t^2x^2+t^2-x^2-1\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t}{4\left(t^2x^2+t^2-x^2-1\right)} + \frac{\sqrt{x^2+1}x}{4\left(t^2x^2+t^2-x^2-1\right)}$$

$$\begin{aligned}\sigma = {} & \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial}{\partial t}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} + \frac{\sqrt{x^2+1}t^2x^2\frac{\partial}{\partial x}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tx^2F}{4\left(t^2x^2F+t^2F-x^2F-F\right)} - \frac{\sqrt{x^2+1}t^2xF}{4\left(t^2x^2F+t^2F-x^2F-F\right)} \\ & + \frac{\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial}{\partial t}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial}{\partial t}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} + \frac{\sqrt{x^2+1}t^2\frac{\partial}{\partial x}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} - \frac{\sqrt{x^2+1}x^2\frac{\partial}{\partial x}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} \\ & - \frac{\sqrt{t+1}\sqrt{t-1}tF}{4\left(t^2x^2F+t^2F-x^2F-F\right)} + \frac{\sqrt{x^2+1}xF}{4\left(t^2x^2F+t^2F-x^2F-F\right)} - \frac{\sqrt{t+1}\sqrt{t-1}\frac{\partial}{\partial t}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)} - \frac{\sqrt{x^2+1}\frac{\partial}{\partial x}F}{2\left(t^2x^2F+t^2F-x^2F-F\right)}\end{aligned}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{t+1}\sqrt{t-1}tx^2F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2xF^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tF^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}xF^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)}$$

$$\begin{aligned}\lambda = {} & -\frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{\sqrt{x^2+1}t^2x^2F\frac{\partial}{\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{\sqrt{t+1}\sqrt{t-1}tx^2F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2xF^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\ & - \frac{\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{\sqrt{t+1}\sqrt{t-1}x^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{\sqrt{x^2+1}t^2F\frac{\partial}{\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} - \frac{\sqrt{x^2+1}x^2F\frac{\partial}{\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} \\ & + \frac{\sqrt{t+1}\sqrt{t-1}tF^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}xF^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} - \frac{\sqrt{x^2+1}F\frac{\partial}{\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}\end{aligned}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{x^2 \frac{\partial}{\partial x} F}{4\sqrt{x^2+1}F} - \frac{t^2 \frac{\partial}{\partial t} F}{\sqrt{t+1}\sqrt{t-1}F} + \frac{\frac{\partial}{\partial x} F}{4\sqrt{x^2+1}F} + \frac{\frac{\partial}{\partial t} F}{\sqrt{t+1}\sqrt{t-1}F}$$

$$\gamma = \frac{\sqrt{t+1}\sqrt{t-1}t^2 F \frac{\partial}{\partial t} F}{a^2t^2 - a^2} + \frac{\sqrt{x^2+1}t^2 F \frac{\partial}{\partial x} F}{2\left(a^2t^2 - a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tF^2}{2\left(a^2t^2 - a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}F \frac{\partial}{\partial t} F}{a^2t^2 - a^2} - \frac{\sqrt{x^2+1}F \frac{\partial}{\partial x} F}{2\left(a^2t^2 - a^2\right)}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{t^2 \frac{\partial^2}{(\partial t)^2} F}{8\,F} + \frac{x^2 \frac{\partial^2}{(\partial x)^2} F}{8\,F} + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}\frac{\partial^2}{\partial t \partial x} F}{2\,F} - \frac{\frac{\partial^2}{(\partial t)^2} F}{8\,F} + \frac{\frac{\partial^2}{(\partial x)^2} F}{8\,F}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{t^2 F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{x^2 F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2} + \frac{F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{t^2 F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{x^2 F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2} - \frac{F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{t^2 F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{x^2 F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2} + \frac{F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{t^2 F^3 \frac{\partial^2}{(\partial t)^2} F}{2\,a^4} + \frac{x^2 F^3 \frac{\partial^2}{(\partial x)^2} F}{2\,a^4} - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^3 \frac{\partial^2}{\partial t \partial x} F}{a^4} - \frac{F^3 \frac{\partial^2}{(\partial t)^2} F}{2\,a^4} + \frac{F^3 \frac{\partial^2}{(\partial x)^2} F}{2\,a^4}$$

$$\Lambda = -\frac{t^2 \frac{\partial}{\partial t} F^2}{2\,a^2} + \frac{t^2 F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{x^2 \frac{\partial}{\partial x} F^2}{2\,a^2} + \frac{x^2 F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2} + \frac{t F \frac{\partial}{\partial t} F}{2\,a^2} + \frac{x F \frac{\partial}{\partial x} F}{2\,a^2} - \frac{F^2}{6\,a^2} + \frac{\frac{\partial}{\partial t} F^2}{2\,a^2} - \frac{F \frac{\partial^2}{(\partial t)^2} F}{4\,a^2} - \frac{\frac{\partial}{\partial x} F^2}{2\,a^2} + \frac{F \frac{\partial^2}{(\partial x)^2} F}{4\,a^2}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{1}{12}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{F^2}{2\,a^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{F^4}{3\,a^4}$$

Newman-Penrose Equations

$$\begin{aligned}
NP1 = {} & \frac{t^4 x^2 \frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{t^4 x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} + \frac{t^4 \frac{\partial}{\partial t} F^2}{4\,(t^2-1)F^2} + \frac{t^4 \frac{\partial^2}{(\partial t)^2} F}{8\,(t^2-1)F} + \frac{t^2 x^2 \frac{\partial}{\partial x} F^2}{4\,(t^2-1)F^2} + \frac{t^2 x^2 \frac{\partial^2}{(\partial x)^2} F}{8\,(t^2-1)F} \\
& + \frac{t^4 x \frac{\partial}{\partial t} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{t^3 x^2 \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{3\,t^3 \frac{\partial}{\partial t} F}{4\,(t^2-1)F} - \frac{3\,t^2 x \frac{\partial}{\partial x} F}{8\,(t^2-1)F} + \frac{t^4 \frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} \\
& - \frac{t^2 x^2 \frac{\partial^2}{\partial t \partial x} F}{(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{t^4 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} - \frac{t^2 x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} + \frac{t^2}{4\,(t^2-1)} - \frac{t^2 \frac{\partial}{\partial t} F^2}{2\,(t^2-1)F^2} \\
& - \frac{t^2 \frac{\partial^2}{(\partial t)^2} F}{4\,(t^2-1)F} + \frac{t^2 \frac{\partial}{\partial x} F^2}{4\,(t^2-1)F^2} - \frac{x^2 \frac{\partial}{\partial x} F^2}{4\,(t^2-1)F^2} + \frac{t^2 \frac{\partial^2}{(\partial x)^2} F}{8\,(t^2-1)F} - \frac{x^2 \frac{\partial^2}{(\partial x)^2} F}{8\,(t^2-1)F} - \frac{t^2 x \frac{\partial}{\partial t} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{t^3 \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} \\
& + \frac{t x^2 \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{3\,t \frac{\partial}{\partial t} F}{4\,(t^2-1)F} + \frac{3\,x \frac{\partial}{\partial x} F}{8\,(t^2-1)F} - \frac{t^2 \frac{\partial^2}{\partial t \partial x} F}{(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{x^2 \frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} \\
& - \frac{t^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} + \frac{x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} + \frac{\frac{\partial}{\partial t} F^2}{4\,(t^2-1)F^2} + \frac{\frac{\partial^2}{(\partial t)^2} F}{8\,(t^2-1)F} - \frac{\frac{\partial}{\partial x} F^2}{4\,(t^2-1)F^2} - \frac{\frac{\partial^2}{(\partial x)^2} F}{8\,(t^2-1)F} \\
& + \frac{x \frac{\partial}{\partial t} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{t \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{\frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{\frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2}
\end{aligned}$$

$$
\begin{aligned}
NP2 = & -\frac{t^4 x^2 \frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{t^4 x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} - \frac{3\,t^4 \frac{\partial}{\partial t} F^2}{4\,(t^2-1)F^2} - \frac{t^4 \frac{\partial^2}{(\partial t)^2} F}{4\,(t^2-1)F} + \frac{t^2 x^2 \frac{\partial}{\partial x} F^2}{2\,(t^2-1)F^2} - \frac{t^2 x^2 \frac{\partial^2}{(\partial x)^2} F}{4\,(t^2-1)F} \\
& + \frac{t^4 x \frac{\partial}{\partial t} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{t^3 x^2 \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{t^3 \frac{\partial}{\partial t} F}{2\,(t^2-1)F} - \frac{5\,t^2 x \frac{\partial}{\partial x} F}{8\,(t^2-1)F} - \frac{t^4 \frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} \\
& + \frac{t^2 x^2 \frac{\partial^2}{\partial t \partial x} F}{(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{t^4 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} + \frac{t^2 x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} - \frac{t^2}{12\,(t^2-1)} + \frac{3\,t^2 \frac{\partial}{\partial t} F^2}{2\,(t^2-1)F^2} \\
& + \frac{t^2 \frac{\partial^2}{(\partial t)^2} F}{2\,(t^2-1)F} + \frac{t^2 \frac{\partial}{\partial x} F^2}{2\,(t^2-1)F^2} - \frac{x^2 \frac{\partial}{\partial x} F^2}{2\,(t^2-1)F^2} - \frac{t^2 \frac{\partial^2}{(\partial x)^2} F}{4\,(t^2-1)F} + \frac{x^2 \frac{\partial^2}{(\partial x)^2} F}{4\,(t^2-1)F} - \frac{t^2 x \frac{\partial}{\partial t} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} + \frac{t^3 \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} \\
& - \frac{t x^2 \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{t \frac{\partial}{\partial t} F}{2\,(t^2-1)F} + \frac{5\,x \frac{\partial}{\partial x} F}{8\,(t^2-1)F} + \frac{t^2 \frac{\partial^2}{\partial t \partial x} F}{(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{x^2 \frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} \\
& + \frac{t^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} - \frac{x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2} - \frac{1}{6\,(t^2-1)} - \frac{3\,\frac{\partial}{\partial t} F^2}{4\,(t^2-1)F^2} - \frac{\frac{\partial^2}{(\partial t)^2} F}{4\,(t^2-1)F} - \frac{\frac{\partial}{\partial x} F^2}{2\,(t^2-1)F^2} + \frac{\frac{\partial^2}{(\partial x)^2} F}{4\,(t^2-1)F} \\
& + \frac{x \frac{\partial}{\partial t} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{t \frac{\partial}{\partial x} F}{8\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{\frac{\partial^2}{\partial t \partial x} F}{2\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F} - \frac{\frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{4\,(t^2-1)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2}
\end{aligned}
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\begin{aligned}
NP6 = & \frac{t^4 x^2 F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{t^4 x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{5\,t^4 \frac{\partial}{\partial t} F^2}{a^2t^2-a^2} - \frac{3\,t^4 F \frac{\partial^2}{(\partial t)^2} F}{2\,(a^2t^2-a^2)} - \frac{t^2 x^2 F \frac{\partial^2}{(\partial x)^2} F}{a^2t^2-a^2} + \frac{t^3 x^2 F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{7\,t^3 F \frac{\partial}{\partial t} F}{2\,(a^2t^2-a^2)} \\
& - \frac{t^2 x F \frac{\partial}{\partial x} F}{a^2t^2-a^2} + \frac{t^4 F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 x^2 F \frac{\partial^2}{\partial t \partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{t^4 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{2\,t^2 x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} \\
& - \frac{t^2 F^2}{3\,(a^2t^2-a^2)} - \frac{10\,t^2 \frac{\partial}{\partial t} F^2}{a^2t^2-a^2} + \frac{3\,t^2 F \frac{\partial^2}{(\partial t)^2} F}{a^2t^2-a^2} - \frac{t^2 F \frac{\partial^2}{(\partial x)^2} F}{a^2t^2-a^2} + \frac{x^2 F \frac{\partial^2}{(\partial x)^2} F}{a^2t^2-a^2} + \frac{t^3 F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{t x^2 F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{7\,t F \frac{\partial}{\partial t} F}{2\,(a^2t^2-a^2)} \\
& + \frac{x F \frac{\partial}{\partial x} F}{a^2t^2-a^2} - \frac{t^2 F \frac{\partial^2}{\partial t \partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{x^2 F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{2\,t^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{x^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{F^2}{12\,(a^2t^2-a^2)} + \frac{5\,\frac{\partial}{\partial t} F^2}{a^2t^2-a^2} - \frac{3\,F \frac{\partial^2}{(\partial t)^2} F}{2\,(a^2t^2-a^2)} + \frac{F \frac{\partial^2}{(\partial x)^2} F}{a^2t^2-a^2} - \frac{t F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{\frac{\partial}{\partial t} F \frac{\partial}{\partial x} F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}}
\end{aligned}
$$

$$
\begin{aligned}
NP7 = {} & \frac{5\,t^4x^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{3\,t^4\frac{\partial}{\partial t}F^2}{2\,(a^2t^2-a^2)} + \frac{t^4F\frac{\partial^2}{(\partial t)^2}F}{4\,(a^2t^2-a^2)} - \frac{t^2x^2\frac{\partial}{\partial x}F^2}{a^2t^2-a^2} - \frac{3\,t^2x^2F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2-a^2)} - \frac{t^4xF\frac{\partial}{\partial t}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} \\
& - \frac{3\,t^3x^2F\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{t^3F\frac{\partial}{\partial t}F}{2\,(a^2t^2-a^2)} - \frac{t^2xF\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)} + \frac{5\,t^4\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{5\,t^2x^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{t^2F^2}{2\,(a^2t^2-a^2)} \\
& + \frac{3\,t^2\frac{\partial}{\partial t}F^2}{a^2t^2-a^2} - \frac{t^2F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2-a^2)} - \frac{t^2\frac{\partial}{\partial x}F^2}{a^2t^2-a^2} + \frac{x^2\frac{\partial}{\partial x}F^2}{a^2t^2-a^2} - \frac{3\,t^2F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2-a^2)} + \frac{3\,x^2F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2-a^2)} + \frac{t^2xF\frac{\partial}{\partial t}F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{3\,t^3F\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{3\,tx^2F\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{tF\frac{\partial}{\partial t}F}{2\,(a^2t^2-a^2)} + \frac{xF\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)} - \frac{5\,t^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{5\,x^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{3\,\frac{\partial}{\partial t}F^2}{2\,(a^2t^2-a^2)} \\
& + \frac{F\frac{\partial^2}{(\partial t)^2}F}{4\,(a^2t^2-a^2)} + \frac{\frac{\partial}{\partial x}F^2}{a^2t^2-a^2} + \frac{3\,F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2-a^2)} - \frac{xF\frac{\partial}{\partial t}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{3\,tF\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{5\,\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}}
\end{aligned}
$$

$$
\begin{aligned}
NP8 = {} & -\frac{3\,t^4\frac{\partial}{\partial t}F^2}{2\,(a^2t^2-a^2)} + \frac{t^4F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2-a^2)} - \frac{t^2x^2\frac{\partial}{\partial x}F^2}{2\,(a^2t^2-a^2)} + \frac{t^2x^2F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2-a^2)} - \frac{t^4xF\frac{\partial}{\partial t}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{3\,t^3x^2F\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{t^3F\frac{\partial}{\partial t}F}{a^2t^2-a^2} + \frac{5\,t^2xF\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)} - \frac{5\,t^2F^2}{6\,(a^2t^2-a^2)} + \frac{3\,t^2\frac{\partial}{\partial t}F^2}{a^2t^2-a^2} - \frac{t^2F\frac{\partial^2}{(\partial t)^2}F}{a^2t^2-a^2} - \frac{t^2\frac{\partial}{\partial x}F^2}{2\,(a^2t^2-a^2)} + \frac{x^2\frac{\partial}{\partial x}F^2}{2\,(a^2t^2-a^2)} + \frac{t^2F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2-a^2)} - \frac{x^2F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2-a^2)} \\
& + \frac{t^2xF\frac{\partial}{\partial t}F}{(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} + \frac{3\,t^3F\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{3\,tx^2F\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{tF\frac{\partial}{\partial t}F}{a^2t^2-a^2} - \frac{5\,xF\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)} \\
& + \frac{F^2}{3\,(a^2t^2-a^2)} - \frac{3\,\frac{\partial}{\partial t}F^2}{2\,(a^2t^2-a^2)} + \frac{F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2-a^2)} + \frac{\frac{\partial}{\partial x}F^2}{2\,(a^2t^2-a^2)} - \frac{F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2-a^2)} - \frac{xF\frac{\partial}{\partial t}F}{2\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}} - \frac{3\,tF\frac{\partial}{\partial x}F}{4\,(a^2t^2-a^2)\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}}
\end{aligned}
$$

$$
\begin{aligned}
NP10 = {} & \frac{3\,t^4x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^4x^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^2x^4F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2x^2F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{3\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{3\,t^3x^2F^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\,t^2x^3F^3\frac{\partial}{\partial x}F}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2xF^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{5\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tx^2F^3\frac{\partial}{\partial x}F}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} + \frac{5\,t^2x^2F^4}{6\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} \\
& + \frac{3\,t^4F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{6\,t^2x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^4F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\,t^2x^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\,t^2x^2F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{x^4F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}txF^4}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}x^2F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{3\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{3\,t^3F^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\,tx^2F^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{3\,t^2xF^3\frac{\partial}{\partial x}F}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} - \frac{3\,x^3F^3\frac{\partial}{\partial x}F}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} - \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}xF^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{5\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tF^3\frac{\partial}{\partial x}F}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} + \frac{5\,t^2F^4}{6\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} \\
& + \frac{2\,x^2F^4}{3\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} - \frac{6\,t^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\,x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\,t^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{x^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{t^2F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\,x^2F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\,tF^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{3\,xF^3\frac{\partial}{\partial x}F}{2\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} + \frac{2\,F^4}{3\,(a^4t^2x^2+a^4t^2-a^4x^2-a^4)} + \frac{3\,F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4}
\end{aligned}
$$

$$
NP12 = \frac{t^2F\frac{\partial^2}{(\partial t)^2}F}{2\,a^2} - \frac{x^2\frac{\partial}{\partial x}F^2}{a^2} + \frac{xF\frac{\partial}{\partial x}F}{a^2} + \frac{F^2}{3\,a^2} - \frac{F\frac{\partial^2}{(\partial t)^2}F}{2\,a^2} - \frac{\frac{\partial}{\partial x}F^2}{a^2}
$$

$$
\begin{aligned}
NP14 = {} & \frac{t^4x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^4x^2F^3\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{t^2x^4F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^2x^4F^3\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2x^2F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{3\,t^3x^2F^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{t^2x^3F^3\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2xF^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{3\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tx^2F^3\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{3\,t^2x^2F^4}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{t^4F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\,t^2x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^4F^3\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& - \frac{t^2x^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\,t^2x^2F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{x^4F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^2x^2F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{x^4F^3\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}txF^4}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}x^2F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{3\,t^3F^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\,tx^2F^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{t^2xF^3\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{x^3F^3\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}xF^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tF^3\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{3\,t^2F^4}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{2\,t^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{t^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{x^2F^3\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{t^2F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\,x^2F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{t^2F^3\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& - \frac{x^2F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\,tF^3\frac{\partial}{\partial t}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{xF^3\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{F^3\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{F^3\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)}
\end{aligned}
$$

$$
\begin{aligned}
NP16 = & -\frac{3\,t^4x^2\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{t^4x^2F\frac{\partial^2}{(\partial t)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{3\,t^2x^4F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{3\,\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& +\frac{t^3x^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}-\frac{3\,t^2x^3F\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2xF\frac{\partial}{\partial t}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tx^2F\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{t^2x^2F^2}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& -\frac{3\,t^4\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,t^2x^2\frac{\partial}{\partial t}F^2}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}+\frac{t^4F\frac{\partial^2}{(\partial t)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{t^2x^2F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{3\,t^2x^2F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& +\frac{3\,x^4F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}txF^2}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{3\,\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& +\frac{t^3F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}-\frac{tx^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}-\frac{3\,t^2xF\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,x^3F\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}xF\frac{\partial}{\partial t}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& +\frac{3\,\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tF\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{t^2F^2}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,t^2\frac{\partial}{\partial t}F^2}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}-\frac{3\,x^2\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{t^2F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& +\frac{x^2F\frac{\partial^2}{(\partial t)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{3\,t^2F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,x^2F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& -\frac{tF\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2}+\frac{3\,xF\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}-\frac{3\,\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{F\frac{\partial^2}{(\partial t)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}+\frac{3\,F\frac{\partial^2}{(\partial x)^2}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}
\end{aligned}
$$

$$
\begin{aligned}
NP17 = {} & \frac{3\,t^4x^2\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{t^4x^2F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{t^2x^4\frac{\partial}{\partial x}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{t^2x^4F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{t^3x^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} \\
& - \frac{3\,t^2x^3F\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}t^2xF\frac{\partial}{\partial t}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tx^2F\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{7\,t^2x^2F^2}{12\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& + \frac{3\,t^4\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{3\,t^2x^2\frac{\partial}{\partial t}F^2}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} - \frac{t^4F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{t^2x^2F\frac{\partial^2}{(\partial t)^2}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{t^2x^2\frac{\partial}{\partial x}F^2}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} \\
& - \frac{x^4\frac{\partial}{\partial x}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{t^2x^2F\frac{\partial^2}{(\partial x)^2}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{x^4F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}txF^2}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{t^3F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} \\
& + \frac{tx^2F\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} - \frac{3\,t^2xF\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{3\,x^3F\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}xF\frac{\partial}{\partial t}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& + \frac{\sqrt{x^2+1}\sqrt{t+1}\sqrt{t-1}tF\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{7\,t^2F^2}{12\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{x^2F^2}{3\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{3\,t^2\frac{\partial}{\partial t}F^2}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} \\
& + \frac{3\,x^2\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{t^2F\frac{\partial^2}{(\partial t)^2}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} - \frac{x^2F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{t^2\frac{\partial}{\partial x}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{x^2\frac{\partial}{\partial x}F^2}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} \\
& - \frac{t^2F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{x^2F\frac{\partial^2}{(\partial x)^2}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{tF\frac{\partial}{\partial t}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{3\,xF\frac{\partial}{\partial x}F}{4\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{F^2}{3\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} \\
& + \frac{3\,\frac{\partial}{\partial t}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{F\frac{\partial^2}{(\partial t)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} - \frac{\frac{\partial}{\partial x}F^2}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)} + \frac{F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^2t^2x^2+a^2t^2-a^2x^2-a^2)}
\end{aligned}
$$

Equations are vanished.

Bianchi Identities

$$
\begin{aligned}
BI2 = {} & \frac{t^4 x^2 \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{t^4 x^2 F \frac{\partial^3}{(\partial t)^3} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{t^2 x^4 F \frac{\partial^3}{\partial t(\partial x)^2} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 x^4 \frac{\partial^2}{\partial t \partial x} F \frac{\partial}{\partial x} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{\sqrt{x^2+1}\, t^2 x^2 F \frac{\partial^3}{(\partial t)^2 \partial x} F}{8\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, t^2 x^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2x^2+a^2)} \\
& + \frac{\sqrt{x^2+1}\, t^2 x^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial x} F}{4\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, x^4 \frac{\partial}{\partial x} F \frac{\partial^2}{(\partial x)^2} F}{4\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, x^4 F \frac{\partial^3}{(\partial x)^3} F}{8\,(a^2x^2+a^2)} + \frac{t^2 x^3 F \frac{\partial^2}{\partial t \partial x} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{\sqrt{x^2+1}\, t x^2 F \frac{\partial^2}{\partial t \partial x} F}{4\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, x^3 F \frac{\partial^2}{(\partial x)^2} F}{2\,(a^2x^2+a^2)} \\
& - \frac{13\, t^2 x^2 F \frac{\partial}{\partial t} F}{12\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{t^4 \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 x^2 \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} F}{(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{t^4 F \frac{\partial^3}{(\partial t)^3} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 x^2 F \frac{\partial^3}{(\partial t)^3} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{t^2 x^2 F \frac{\partial^3}{\partial t(\partial x)^2} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{x^4 F \frac{\partial^3}{\partial t(\partial x)^2} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 x^2 \frac{\partial^2}{\partial t \partial x} F \frac{\partial}{\partial x} F}{(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{x^4 \frac{\partial^2}{\partial t \partial x} F \frac{\partial}{\partial x} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{\sqrt{x^2+1}\, t^2 F \frac{\partial^3}{(\partial t)^2 \partial x} F}{8\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, x^2 F \frac{\partial^3}{(\partial t)^2 \partial x} F}{8\,(a^2x^2+a^2)} \\
& + \frac{\sqrt{x^2+1}\, t^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, x^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2x^2+a^2)} - \frac{11\sqrt{x^2+1}\, x^2 F \frac{\partial}{\partial x} F}{12\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, t^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial x} F}{4\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, x^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial x} F}{4\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, x^2 \frac{\partial}{\partial x} F \frac{\partial^2}{(\partial x)^2} F}{2\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, x^2 F \frac{\partial^3}{(\partial x)^3} F}{4\,(a^2x^2+a^2)} \\
& + \frac{t x^2 F^2}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{t^2 x F \frac{\partial^2}{\partial t \partial x} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{x^3 F \frac{\partial^2}{\partial t \partial x} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{\sqrt{x^2+1}\, x F^2}{3\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, t F \frac{\partial^2}{\partial t \partial x} F}{4\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, x F \frac{\partial^2}{(\partial x)^2} F}{2\,(a^2x^2+a^2)} \\
& - \frac{13\, t^2 F \frac{\partial}{\partial t} F}{12\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{13\, x^2 F \frac{\partial}{\partial t} F}{12\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} F}{(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{x^2 \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 F \frac{\partial^3}{(\partial t)^3} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{x^2 F \frac{\partial^3}{(\partial t)^3} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{t^2 F \frac{\partial^3}{\partial t(\partial x)^2} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{x^2 F \frac{\partial^3}{\partial t(\partial x)^2} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2 \frac{\partial^2}{\partial t \partial x} F \frac{\partial}{\partial x} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{x^2 \frac{\partial^2}{\partial t \partial x} F \frac{\partial}{\partial x} F}{(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{\sqrt{x^2+1}\, F \frac{\partial^3}{(\partial t)^2 \partial x} F}{8\,(a^2x^2+a^2)} \\
& - \frac{\sqrt{x^2+1}\, \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial x} F}{2\,(a^2x^2+a^2)} - \frac{11\sqrt{x^2+1}\, F \frac{\partial}{\partial x} F}{12\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial x} F}{4\,(a^2x^2+a^2)} - \frac{\sqrt{x^2+1}\, \frac{\partial}{\partial x} F \frac{\partial^2}{(\partial x)^2} F}{4\,(a^2x^2+a^2)} + \frac{\sqrt{x^2+1}\, F \frac{\partial^3}{(\partial x)^3} F}{8\,(a^2x^2+a^2)} + \frac{t F^2}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{x F \frac{\partial^2}{\partial t \partial x} F}{4\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{13\, F \frac{\partial}{\partial t} F}{12\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{\frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{F \frac{\partial^3}{(\partial t)^3} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} - \frac{F \frac{\partial^3}{\partial t(\partial x)^2} F}{8\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}} + \frac{\frac{\partial^2}{\partial t \partial x} F \frac{\partial}{\partial x} F}{2\,(a^2x^2+a^2)\sqrt{t+1}\sqrt{t-1}}
\end{aligned}
$$

$$
\begin{aligned}
BI3 = & -\frac{t^4x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^4x^2F^3\frac{\partial^3}{(\partial t)^3}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2x^4F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{t^2x^4F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{\sqrt{x^2+1}t^2x^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\,(a^4x^2+a^4)} \\
& + \frac{\sqrt{x^2+1}t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4x^2+a^4} + \frac{\sqrt{x^2+1}t^2x^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}x^4F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^4x^2+a^4)} + \frac{\sqrt{x^2+1}x^4F^3\frac{\partial^3}{(\partial x)^3}F}{4\,(a^4x^2+a^4)} - \frac{t^2x^3F^3\frac{\partial^2}{\partial t\partial x}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& - \frac{\sqrt{x^2+1}tx^2F^3\frac{\partial^2}{\partial t\partial x}F}{2\,(a^4x^2+a^4)} + \frac{\sqrt{x^2+1}x^3F^3\frac{\partial^2}{(\partial x)^2}F}{a^4x^2+a^4} + \frac{13\,t^2x^2F^3\frac{\partial}{\partial t}F}{6\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^4F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{2\,t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^4F^3\frac{\partial^3}{(\partial t)^3}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{t^2x^2F^3\frac{\partial^3}{(\partial t)^3}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2x^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{x^4F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{2\,t^2x^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{x^4F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{\sqrt{x^2+1}t^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}x^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\,(a^4x^2+a^4)} + \frac{\sqrt{x^2+1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4x^2+a^4} - \frac{\sqrt{x^2+1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4x^2+a^4} - \frac{5\sqrt{x^2+1}x^2F^3\frac{\partial}{\partial x}F}{2\,(a^4x^2+a^4)} + \frac{\sqrt{x^2+1}t^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\,(a^4x^2+a^4)} \\
& - \frac{\sqrt{x^2+1}x^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}x^2F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{a^4x^2+a^4} + \frac{\sqrt{x^2+1}x^2F^3\frac{\partial^3}{(\partial x)^3}F}{2\,(a^4x^2+a^4)} - \frac{2\,tx^2F^4}{3\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2xF^3\frac{\partial^2}{\partial t\partial x}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{x^3F^3\frac{\partial^2}{\partial t\partial x}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{2\sqrt{x^2+1}xF^4}{3\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}tF^3\frac{\partial^2}{\partial t\partial x}F}{2\,(a^4x^2+a^4)} + \frac{\sqrt{x^2+1}xF^3\frac{\partial^2}{(\partial x)^2}F}{a^4x^2+a^4} + \frac{13\,t^2F^3\frac{\partial}{\partial t}F}{6\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{13\,x^2F^3\frac{\partial}{\partial t}F}{6\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{2\,t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{t^2F^3\frac{\partial^3}{(\partial t)^3}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{x^2F^3\frac{\partial^3}{(\partial t)^3}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{t^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& + \frac{x^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{t^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{2\,x^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{\sqrt{x^2+1}F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4x^2+a^4} \\
& - \frac{5\sqrt{x^2+1}F^3\frac{\partial}{\partial x}F}{2\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\,(a^4x^2+a^4)} - \frac{\sqrt{x^2+1}F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\,(a^4x^2+a^4)} + \frac{\sqrt{x^2+1}F^3\frac{\partial^3}{(\partial x)^3}F}{4\,(a^4x^2+a^4)} - \frac{2\,tF^4}{3\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{xF^3\frac{\partial^2}{\partial t\partial x}F}{2\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} \\
& - \frac{13\,F^3\frac{\partial}{\partial t}F}{6\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{F^3\frac{\partial^3}{(\partial t)^3}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} + \frac{F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\,(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}} - \frac{F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{(a^4x^2+a^4)\sqrt{t+1}\sqrt{t-1}}
\end{aligned}
$$

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\begin{aligned}
BI5 = {} & \frac{5\sqrt{t+1}\sqrt{t-1}t^4x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}t^4x^2F\frac{\partial^3}{(\partial t)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^4F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2x^4\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^4\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{x^2+1}t^4x^2F\frac{\partial^3}{(\partial t)^2\partial x}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{7\sqrt{x^2+1}t^4x^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^4x^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{x^2+1}t^2x^4\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2x^4F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^3x^2\frac{\partial}{\partial t}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^3x^2F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^3F\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}t^2x^3\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tx^4F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{7\sqrt{x^2+1}t^3x^2F\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^3x^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}t^2x^3\frac{\partial}{\partial x}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& - \frac{\sqrt{x^2+1}t^2x^3F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F\frac{\partial}{\partial t}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}t^4\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}t^4F\frac{\partial^3}{(\partial t)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2x^2F\frac{\partial^3}{(\partial t)^3}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F\frac{\partial^3}{\partial t(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^4F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} - \frac{3\sqrt{t+1}\sqrt{t-1}x^4\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^4\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& - \frac{5\sqrt{x^2+1}t^4F\frac{\partial^3}{(\partial t)^2\partial x}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{5\sqrt{x^2+1}t^2x^2F\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{7\sqrt{x^2+1}t^4\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{7\sqrt{x^2+1}t^2x^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{5\sqrt{x^2+1}t^2x^2F\frac{\partial}{\partial x}F}{12\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^4\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{x^2+1}t^2x^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{x^2+1}t^2x^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{x^2+1}x^4\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2x^2F\frac{\partial^3}{(\partial x)^3}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}x^4F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tx^2F^2}{3\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^3\frac{\partial}{\partial t}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tx^2\frac{\partial}{\partial t}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^3F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}tx^2F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2xF\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^3F\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}t^2x\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}x^3\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tx^2F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}t^2xF^2}{3\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{7\sqrt{x^2+1}t^3F\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{7\sqrt{x^2+1}tx^2F\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& - \frac{\sqrt{x^2+1}t^3\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}tx^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}t^2x\frac{\partial}{\partial x}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}x^3\frac{\partial}{\partial x}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2xF\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}x^3F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial}{\partial t}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2F\frac{\partial}{\partial t}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial^3}{(\partial t)^3}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}x^2F\frac{\partial^3}{(\partial t)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2F\frac{\partial^3}{\partial t(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{5\sqrt{x^2+1}t^2F\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{x^2+1}x^2F\frac{\partial^3}{(\partial t)^2\partial x}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{7\sqrt{x^2+1}t^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^2t^2x^2+a^2t^2-a^2x^2-a^2} + \frac{7\sqrt{x^2+1}x^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{5\sqrt{x^2+1}t^2F\frac{\partial}{\partial x}F}{12\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{x^2+1}x^2F\frac{\partial}{\partial x}F}{12\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{\sqrt{x^2+1}t^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}x^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{3\sqrt{x^2+1}t^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{x^2+1}x^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}x^2F\frac{\partial^3}{(\partial x)^3}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}tF^2}{3\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}t\frac{\partial}{\partial t}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}tF\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}xF\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}x\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tF\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& - \frac{\sqrt{x^2+1}xF^2}{3\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{7\sqrt{x^2+1}tF\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}t\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}x\frac{\partial}{\partial x}F^2}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}xF\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}F\frac{\partial}{\partial t}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{5\sqrt{t+1}\sqrt{t-1}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}F\frac{\partial^3}{(\partial t)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{x^2+1}F\frac{\partial^3}{(\partial t)^2\partial x}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} \\
& + \frac{7\sqrt{x^2+1}\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{2\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{5\sqrt{x^2+1}F\frac{\partial}{\partial x}F}{12\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{\sqrt{x^2+1}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} - \frac{3\sqrt{x^2+1}\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)} + \frac{\sqrt{x^2+1}F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2x^2+a^2t^2-a^2x^2-a^2\right)}
\end{aligned}
$$

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\begin{aligned}
BI6 = {} & \frac{5\sqrt{t+1}\sqrt{t-1}t^4x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}t^4x^2F^3\frac{\partial^3}{(\partial t)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^4F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{7\sqrt{t+1}\sqrt{t-1}t^2x^4F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^4F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{5\sqrt{x^2+1}t^4x^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{7\sqrt{x^2+1}t^4x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}t^4x^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{5\sqrt{x^2+1}t^2x^4F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{x^2+1}t^2x^4F^3\frac{\partial^3}{(\partial x)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^3x^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{2\sqrt{t+1}\sqrt{t-1}t^3x^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^3F^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}t^2x^3F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{3\sqrt{x^2+1}t^3x^2F^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{x^2+1}t^3x^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}t^2x^3F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{\sqrt{x^2+1}t^2x^3F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F^3\frac{\partial}{\partial t}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}t^4F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{3\sqrt{t+1}\sqrt{t-1}t^4F^3\frac{\partial^3}{(\partial t)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2x^2F^3\frac{\partial^3}{(\partial t)^3}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^4F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{14\sqrt{t+1}\sqrt{t-1}t^2x^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{7\sqrt{t+1}\sqrt{t-1}x^4F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}x^4F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{x^2+1}t^4F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{5\sqrt{x^2+1}t^2x^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{7\sqrt{x^2+1}t^4F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{14\sqrt{x^2+1}t^2x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{5\sqrt{x^2+1}t^2x^2F^3\frac{\partial}{\partial x}F}{6\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& - \frac{\sqrt{x^2+1}t^4F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{x^2+1}t^2x^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{5\sqrt{x^2+1}t^2x^2F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{5\sqrt{x^2+1}x^4F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{x^2+1}t^2x^2F^3\frac{\partial^3}{(\partial x)^3}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{x^2+1}x^4F^3\frac{\partial^3}{(\partial x)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& - \frac{2\sqrt{t+1}\sqrt{t-1}tx^2F^4}{3\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^3F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{t+1}\sqrt{t-1}tx^2F^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\sqrt{t+1}\sqrt{t-1}t^3F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{t+1}\sqrt{t-1}tx^2F^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{t+1}\sqrt{t-1}t^2xF^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}x^3F^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}t^2xF^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{t+1}\sqrt{t-1}x^3F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{2\sqrt{x^2+1}t^2xF^4}{3\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{3\sqrt{x^2+1}t^3F^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{3\sqrt{x^2+1}tx^2F^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{\sqrt{x^2+1}t^3F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}tx^2F^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}t^2xF^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{x^2+1}x^3F^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{x^2+1}t^2xF^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}x^3F^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2F^3\frac{\partial}{\partial t}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2F^3\frac{\partial}{\partial t}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{5\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^2F^3\frac{\partial^3}{(\partial t)^3}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& - \frac{3\sqrt{t+1}\sqrt{t-1}x^2F^3\frac{\partial^3}{(\partial t)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{7\sqrt{t+1}\sqrt{t-1}t^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{14\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{5\sqrt{x^2+1}t^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{6\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{x^2+1}x^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{14\sqrt{x^2+1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{7\sqrt{x^2+1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{5\sqrt{x^2+1}t^2F^3\frac{\partial}{\partial x}F}{6\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{x^2+1}x^2F^3\frac{\partial}{\partial x}F}{6\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{\sqrt{x^2+1}t^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}x^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{5\sqrt{x^2+1}t^2F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{x^2+1}x^2F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& + \frac{\sqrt{x^2+1}t^2F^3\frac{\partial^3}{(\partial x)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{x^2+1}x^2F^3\frac{\partial^3}{(\partial x)^3}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{2\sqrt{t+1}\sqrt{t-1}tF^4}{3\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tF^2\frac{\partial}{\partial t}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{t+1}\sqrt{t-1}tF^3\frac{\partial^2}{(\partial t)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{t+1}\sqrt{t-1}xF^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}xF^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{2\sqrt{x^2+1}xF^4}{3\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{3\sqrt{x^2+1}tF^3\frac{\partial^2}{\partial t\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{x^2+1}tF^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{\sqrt{x^2+1}xF^2\frac{\partial}{\partial x}F^2}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{x^2+1}xF^3\frac{\partial^2}{(\partial x)^2}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial}{\partial t}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial^3}{(\partial t)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{7\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} - \frac{\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} \\
& + \frac{5\sqrt{x^2+1}F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{7\sqrt{x^2+1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4t^2x^2+a^4t^2-a^4x^2-a^4} + \frac{5\sqrt{x^2+1}F^3\frac{\partial}{\partial x}F}{6\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{x^2+1}F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} + \frac{5\sqrt{x^2+1}F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{2\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)} - \frac{\sqrt{x^2+1}F^3\frac{\partial^3}{(\partial x)^3}F}{4\left(a^4t^2x^2+a^4t^2-a^4x^2-a^4\right)}
\end{aligned}
$$

$$
\begin{aligned}
BI9 = {} & \frac{11\sqrt{t+1}\sqrt{t-1}t^4\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}t^4F\frac{\partial^3}{(\partial t)^3}F}{8\left(a^2t^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} + \frac{2\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^2t^2-a^2} + \frac{\sqrt{t+1}\sqrt{t-1}t^2x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} \\
& - \frac{7\sqrt{x^2+1}t^4F\frac{\partial^3}{(\partial t)^2\partial x}F}{8\left(a^2t^2-a^2\right)} + \frac{4\sqrt{x^2+1}t^4\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^2t^2-a^2} - \frac{5\sqrt{x^2+1}t^4\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{8\left(a^2t^2-a^2\right)} + \frac{9\sqrt{x^2+1}t^2x^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2x^2F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2-a^2\right)} + \frac{3\sqrt{t+1}\sqrt{t-1}t^3\frac{\partial}{\partial t}F^2}{4\left(a^2t^2-a^2\right)} \\
& - \frac{7\sqrt{t+1}\sqrt{t-1}t^3F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2-a^2\right)} + \frac{\sqrt{t+1}\sqrt{t-1}t^2xF\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}t^2x\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tx^2F\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2-a^2\right)} - \frac{7\sqrt{x^2+1}t^3F\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2-a^2\right)} \\
& - \frac{3\sqrt{x^2+1}t^3\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} + \frac{3\sqrt{x^2+1}t^2x\frac{\partial}{\partial x}F^2}{4\left(a^2t^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2xF\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial}{\partial t}F}{4\left(a^2t^2-a^2\right)} - \frac{11\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2-a^2\right)} + \frac{5\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial^3}{(\partial t)^3}F}{4\left(a^2t^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}t^2F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}x^2F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} + \frac{2\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^2t^2-a^2} - \frac{2\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^2t^2-a^2} + \frac{\sqrt{t+1}\sqrt{t-1}t^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}x^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} + \frac{7\sqrt{x^2+1}t^2F\frac{\partial^3}{(\partial t)^2\partial x}F}{4\left(a^2t^2-a^2\right)} - \frac{8\sqrt{x^2+1}t^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^2t^2-a^2} - \frac{\sqrt{x^2+1}t^2F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} + \frac{5\sqrt{x^2+1}t^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} + \frac{9\sqrt{x^2+1}t^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} \\
& - \frac{9\sqrt{x^2+1}x^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{\sqrt{x^2+1}t^2F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2-a^2\right)} + \frac{\sqrt{x^2+1}x^2F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2-a^2\right)} - \frac{3\sqrt{t+1}\sqrt{t-1}t\frac{\partial}{\partial t}F^2}{4\left(a^2t^2-a^2\right)} + \frac{7\sqrt{t+1}\sqrt{t-1}tF\frac{\partial^2}{(\partial t)^2}F}{4\left(a^2t^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}xF\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2-a^2\right)} \\
& + \frac{3\sqrt{t+1}\sqrt{t-1}x\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}tF\frac{\partial^2}{(\partial x)^2}F}{4\left(a^2t^2-a^2\right)} + \frac{7\sqrt{x^2+1}tF\frac{\partial^2}{\partial t\partial x}F}{4\left(a^2t^2-a^2\right)} + \frac{3\sqrt{x^2+1}t\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} - \frac{3\sqrt{x^2+1}x\frac{\partial}{\partial x}F^2}{4\left(a^2t^2-a^2\right)} + \frac{\sqrt{x^2+1}xF\frac{\partial^2}{(\partial x)^2}F}{2\left(a^2t^2-a^2\right)} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}F\frac{\partial}{\partial t}F}{4\left(a^2t^2-a^2\right)} + \frac{11\sqrt{t+1}\sqrt{t-1}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{5\sqrt{t+1}\sqrt{t-1}F\frac{\partial^3}{(\partial t)^3}F}{8\left(a^2t^2-a^2\right)} - \frac{\sqrt{t+1}\sqrt{t-1}F\frac{\partial^3}{\partial t(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{2\sqrt{t+1}\sqrt{t-1}\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^2t^2-a^2} \\
& - \frac{\sqrt{t+1}\sqrt{t-1}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} - \frac{7\sqrt{x^2+1}F\frac{\partial^3}{(\partial t)^2\partial x}F}{8\left(a^2t^2-a^2\right)} + \frac{4\sqrt{x^2+1}\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^2t^2-a^2} + \frac{\sqrt{x^2+1}F\frac{\partial}{\partial x}F}{4\left(a^2t^2-a^2\right)} - \frac{5\sqrt{x^2+1}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{8\left(a^2t^2-a^2\right)} - \frac{9\sqrt{x^2+1}\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{8\left(a^2t^2-a^2\right)} + \frac{\sqrt{x^2+1}F\frac{\partial^3}{(\partial x)^3}F}{8\left(a^2t^2-a^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
BI11 = {} & \frac{11\sqrt{t+1}\sqrt{t-1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\,a^4} - \frac{5\sqrt{t+1}\sqrt{t-1}t^2F^3\frac{\partial^3}{(\partial t)^3}F}{4\,a^4} + \frac{\sqrt{t+1}\sqrt{t-1}x^2F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\,a^4} + \frac{8\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4} \\
& + \frac{\sqrt{t+1}\sqrt{t-1}x^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{4\,a^4} + \frac{7\sqrt{x^2+1}t^2F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\,a^4} - \frac{8\sqrt{x^2+1}t^2F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4} + \frac{\sqrt{x^2+1}t^2F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\,a^4} - \frac{13\sqrt{x^2+1}x^2F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{4\,a^4} \\
& + \frac{\sqrt{x^2+1}x^2F^3\frac{\partial^3}{(\partial x)^3}F}{4\,a^4} + \frac{3\sqrt{t+1}\sqrt{t-1}tF^2\frac{\partial}{\partial t}F^2}{2\,a^4} - \frac{3\sqrt{t+1}\sqrt{t-1}tF^3\frac{\partial^2}{(\partial t)^2}F}{a^4} + \frac{\sqrt{t+1}\sqrt{t-1}xF^3\frac{\partial^2}{\partial t\partial x}F}{2\,a^4} - \frac{3\sqrt{t+1}\sqrt{t-1}xF^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,a^4} \\
& + \frac{3\sqrt{x^2+1}tF^3\frac{\partial^2}{\partial t\partial x}F}{2\,a^4} + \frac{3\sqrt{x^2+1}tF^2\frac{\partial}{\partial t}F\frac{\partial}{\partial x}F}{2\,a^4} - \frac{3\sqrt{x^2+1}xF^2\frac{\partial}{\partial x}F^2}{2\,a^4} + \frac{\sqrt{x^2+1}xF^3\frac{\partial^2}{(\partial x)^2}F}{a^4} - \frac{\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial}{\partial t}F}{2\,a^4} - \frac{11\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}F}{4\,a^4} \\
& + \frac{5\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial^3}{(\partial t)^3}F}{4\,a^4} + \frac{\sqrt{t+1}\sqrt{t-1}F^3\frac{\partial^3}{\partial t(\partial x)^2}F}{4\,a^4} + \frac{8\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial^2}{\partial t\partial x}F\frac{\partial}{\partial x}F}{a^4} + \frac{\sqrt{t+1}\sqrt{t-1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial x)^2}F}{4\,a^4} - \frac{7\sqrt{x^2+1}F^3\frac{\partial^3}{(\partial t)^2\partial x}F}{4\,a^4} \\
& + \frac{8\sqrt{x^2+1}F^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial x}F}{a^4} + \frac{\sqrt{x^2+1}F^3\frac{\partial}{\partial x}F}{2\,a^4} - \frac{\sqrt{x^2+1}F^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial x}F}{4\,a^4} - \frac{13\sqrt{x^2+1}F^2\frac{\partial}{\partial x}F\frac{\partial^2}{(\partial x)^2}F}{4\,a^4} + \frac{\sqrt{x^2+1}F^3\frac{\partial^3}{(\partial x)^3}F}{4\,a^4}
\end{aligned}
$$

Equations are vanished.

Petrov Type

Type "I"